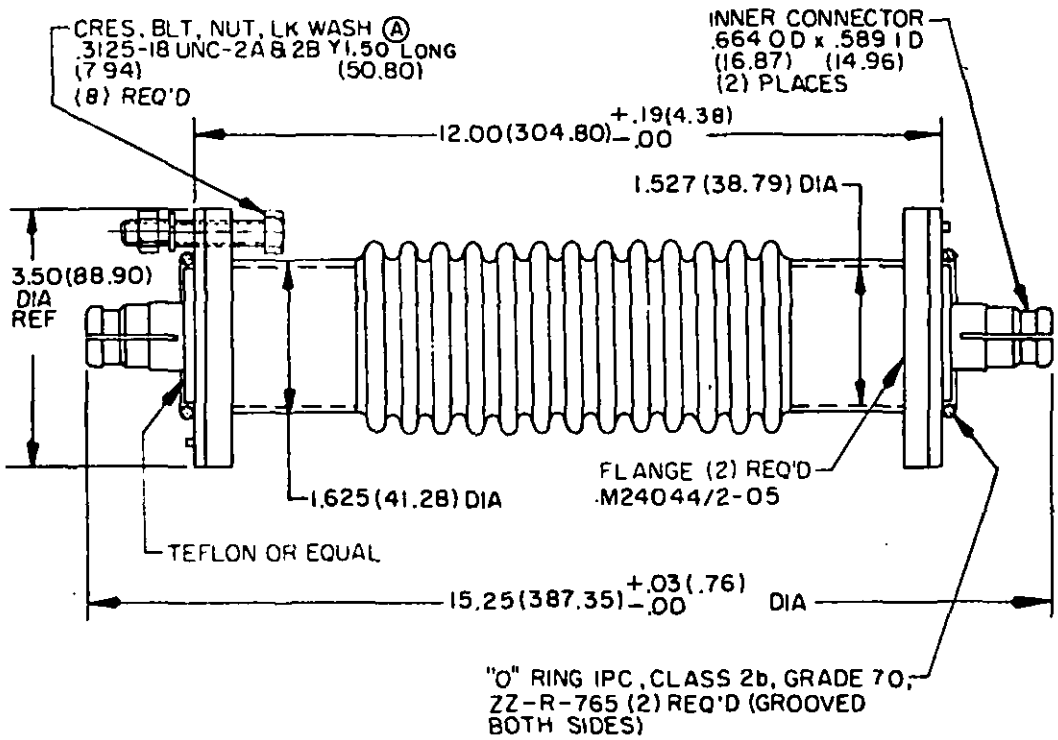
MILITARY SPECIFICATION SHEET

LINE ASSEMBLY, RADIO FREQUENCY TRANSMISSION 1-5/8 INCH STRAIGHT FLEXIBLE SECTIONS, FLANGED

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the line assembly described herein shall consist of this specification and the latest issue of MIL-L-28796.



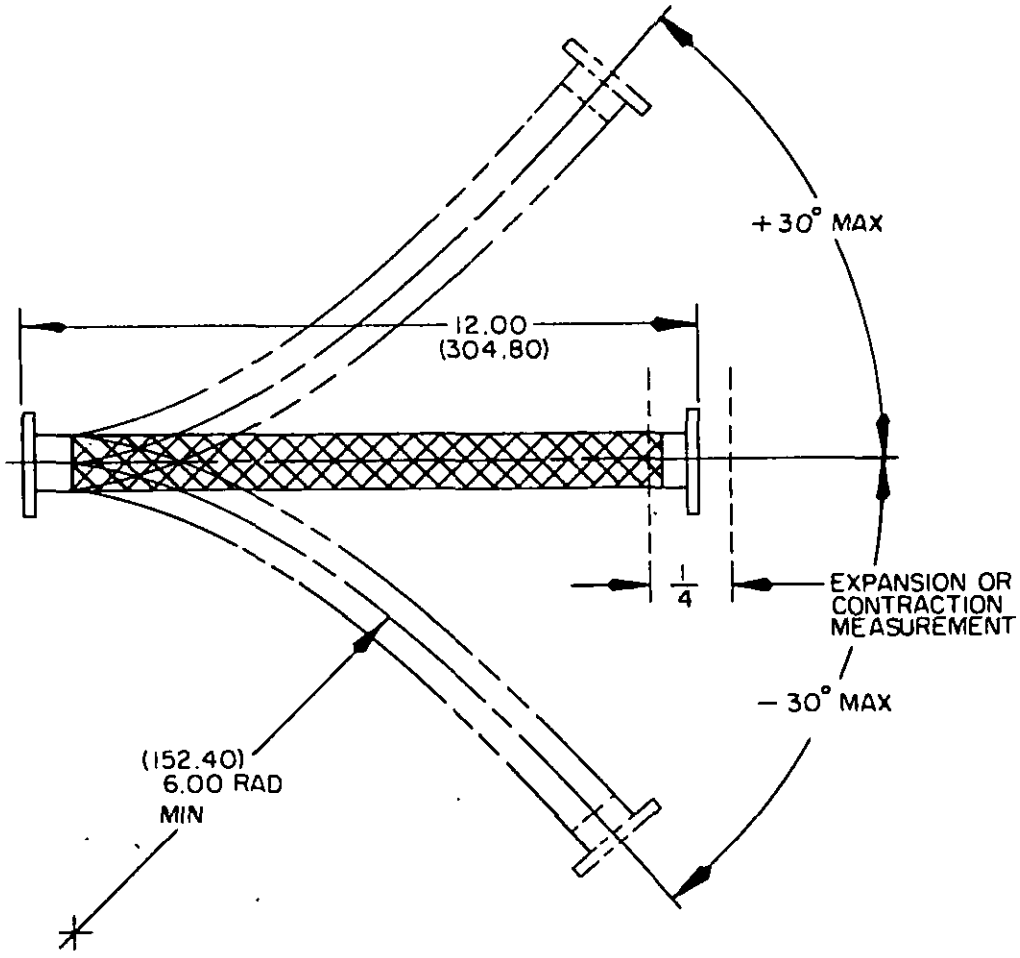
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Metric equivalents are in parentheses.

FIGURE 1. Line assembly.

Ⓐ denotes changes

FSC 5985

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. Metric equivalents are in parentheses.
4. One cycle shall consist of starting with the line assembly streight, bend 30° and return through 0° to 130° in the opposit direction, then return to straight again.

FIGURE 2. Bend test.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Operating frequency range: 10 to 1,600 MHz.

Weight: 5 pounds maximum.

Offset alinement: 0.3 inch maximum.

Transmission line material: Copper alloy.

Insertion loss: 0.02 dB maximum.

VSWR:

1.08 maximum at 100 to 880 MHz.

1.10 maximum at 880 to 1,600 MHz.

Impedance: 50 ±0.3 ohms.

Velocity of propagation: 99.6 percent minimum.

Dielectric withstanding voltage: 6.6 kV, 60 Hz.

Pressurization:

Operating - Gage pressure of 30 ±2 lbf/in^2.

Bend: Mount the line assemblies in a suitable test fixture and flex over a ±30° range at 28 strokes per minute for 2,500 cycles. The expansion and contraction shall be measured, see figure 2. This test shall be performed during first article, group I, and also in group A inspection.

RF power handling capability:

Average - 1.53 kilowatts minimum.

Peak - 140 kilowatts minimum.

Part number: M28796/7-001.

Custodian:
Navy - EC

Review activities:
Navy - SH, OS
DLA - ES

User activities:
Navy - MC, AS

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5985-N491-4)

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